HITAN UWB · DW1000 SS-TWR FIRMWARE

Migration to 110 kbps Two-Way Ranging

Line-exact changelist for the Tag and Anchor firmware, grounded in the Decawave reference driver and DW1000 User Manual.

Target: 850 kbps → 110 kbps **Radio (unchanged):** Ch 2 · PRF 64M · PLEN 1024 · code 9 **Scheme:** Single-sided TWR **Date:** 2026-06-09

> **Why ranging breaks at 110 kbps today.** The anchor computes `clockOffsetRatio` with `FREQ_OFFSET_MULTIPLIER` (the 1024-sample constant). At 110 kbps the carrier integrator accumulates over **8192** samples, so the raw value is **8× larger** — the correction is over-applied, `(1 − clockOffsetRatio)` swings negative, `rtd_resp > rtd_init`, and the time-of-flight goes negative. Removing `abs()` exposes the symptom; fixing the multiplier (change **A5**) is the actual cure.

Line numbers reference `documentation/fw_actual_ref/` (mirror of the source tree). Severity legend: BLOCKER ranging wrong/broken IMPORTANT quality/reliability OPTIONAL choice/verify.

File 1 — Anchor configuration `anc/inc/dw1000.h`

#	LINE	CURRENT	CHANGE TO	SEVERITY
A1	126	`DWT_BR_850K,`	`DWT_BR_110K,` — the data-rate switch.	BLOCKER
A2	128	`1100 //(1024 + 64 - 32)`	`1058 // (1024+1+64−32) 110k 64-sym SFD`. At 110k the SFD is 64 symbols; sized timeout is `PLEN+1+SFD−PAC`. Any value ≥1058 works.	BLOCKER
A3	231	`RESP_RX_TIMEOUT_UUS 20000`	`RESP_RX_TIMEOUT_UUS 25000`. 110k frames are ~8× longer over the air; add multipath/offset headroom.	IMPORTANT
A4	122	`DWT_PAC32,`	*Leave as-is* — valid with PLEN_1024 (UM Table 6). Verify only if preamble is shortened.	VERIFY

Optional (must match Tag): to use Decawave's more-robust 64-symbol SFD instead of standard IEEE, set `nsSFD` at line 125 from `0,` → `1,` (UM Table 21 recommends it for 110k). Both ends must agree.

File 2 — Anchor code `anc/src/dw1000.cpp`

A5 BLOCKER Data-rate-aware carrier-integrator multiplier (lines 296–298 — the root-cause fix)

```
/* was: uses the 1024-sample constant unconditionally */
double freqOffsetMultiplier =
        (config.dataRate == DWT_BR_110K) ? FREQ_OFFSET_MULTIPLIER_110KB
                                         : FREQ_OFFSET_MULTIPLIER;
clockOffsetRatio = dwt_readcarrierintegrator()
        * (freqOffsetMultiplier * HERTZ_TO_PPM_MULTIPLIER_CHAN_2 / 1.0e6);
```

`config` is the file-static `dwt_config_t` from the header and is in scope here. 850k and 6.8M correctly fall through to the 1024 constant — which is why 850k works today.

A6 BLOCKER Validity guard instead of `abs()` (lines 309–312)

After A5 the value is physical, but at 110k you sit closer to the 32-bit timestamp wraparound ceiling (~67 ms). A negative result now means a *bad* sample — reject it rather than fold it into a fake short range.

```
double tof = ((rtd_init - rtd_resp) / 2.0) * DWT_TIME_UNITS;
if (tof < 0) {                          // non-physical: clock/wraparound glitch
    dwmInfo.dw_ready = DW_NO_VALID_DATA;     // drop the sample
} else {
    dwmInfo.tof      = tof;
    dwmInfo.distance = dwmInfo.tof * SPEED_OF_LIGHT * 100;
    dwmInfo.dw_ready = VALID_DATA;
    dwt_readdiagnostics(&dwmInfo.diagnostics);
    dwmInfo.wrong_ID = false;
}
```

Minimal alternative: simply restore `abs()`. Once A5 is in, `abs()` only fires on rare wraparound and no longer masks a calibration bug — but the guard above is better because it discards the glitch instead of reporting a wrong distance.

No change at lines 118–119 (`dwt_setrxaftertxdelay` / `dwt_setrxtimeout`) beyond A3 — they already read the macros. `dwt_configure()` (line 98) auto-applies `SYS_CFG_RXM110K` , `LDE_REPC÷8` , `DRX_TUNE0b` , `DRX_TUNE1b_110K` from the enum.

File 3 — Tag configuration `tag/src/dw1000.cpp`

The Tag's active `config` struct lives in the `.cpp` (lines 40–50), not the header.

#	LINE	CURRENT	CHANGE TO	SEVERITY
T1	47	`DWT_BR_850K,`	`DWT_BR_110K,` — **must match anchor A1** or ranging stops entirely (UM Table 21: RX = TX).	BLOCKER
T2	49	`1100 //(1024 + 1 + 16)`	`1058 // (1024+1+64−32) 110k 64-sym SFD` — match A2.	BLOCKER
T3	46	`0,` (nsSFD)	Set to `1,` *only if* you chose the optional DW SFD on the anchor. SFD type must be identical on both ends.	OPTIONAL
T4	43	`DWT_PAC32,`	*Leave as-is* — valid with PLEN_1024.	VERIFY

File 4 — Tag header `tag/Inc/dw1000.h`

#	LINE	CURRENT	CHANGE TO	SEVERITY
T5	84	`TIMEOUT_PACS 1024/32*2`	`TIMEOUT_PACS (1024/32*4)` . Preamble-detect timeout (PAC units), used at `dw1000.cpp:132 & 284` . At 110k the receiver hunts longer; widen to ~4 preambles. Tune empirically.	IMPORTANT
T6	138	`POLL_RX_TO_RESP_TX_DLY_UUS 2000`	*Keep ~2000.* Contributes to round-trip time; keep total well under ~67 ms (32-bit wraparound limit, see A6). **Do not increase** at 110k without re-checking that margin.	VERIFY

No change to antenna delays (tag 92–93 = 16500/16500; anchor 149–150 = 16550/16500) — antenna delay is data-rate independent (APS014). TX power, AGC, channel/PRF/preamble code all unchanged.

At a glance

MANDATORY

- **A5** — 110k carrier-integrator multiplier *(the fix)*
- **A1 + T1** — `DWT_BR_110K` on both
- **A2 + T2** — 64-sym SFD timeout on both
- **A6** — validity guard / `abs()` decision

IMPORTANT

- **A3** — wider `RESP_RX_TIMEOUT_UUS`
- **T5** — wider `TIMEOUT_PACS`

VERIFY / DECIDE

- **A4 + T4** — PAC stays 32 (OK w/ PLEN1024)
- **T3** + nsSFD — std vs DW 64-sym SFD; must match
- **T6** — keep tag reply delay small

> **Auto-handled by `dwt_configure()` — no code change:** `SYS_CFG_RXM110K` , `LDE_REPC÷8` , `DRX_TUNE0b` , `DRX_TUNE1b_110K` . **Host-side (outside firmware):** `rf_diagnostics.py` hard-codes the 8-symbol SFD adjustment (`SFD_ADJ_STD_8`); at 110k it should be `SFD_ADJ_DW_64` for correct RX/FP *power* readouts. Does not affect distance.

Basis for the 8× factor: the two header macros differ only by their divisor (`1024` vs `8192`), and DW1000 User Manual §7.2.40.11 (DRX_CAR_INT) states $N_{samples}$ = `8192 for 110 kb/s, 1024 otherwise` . Verified against `deca_device_api.h` and `deca_device.c` .